Real-Time Driver and Road Monitoring for Risk Prediction

25-26J-291

Project Proposal Report

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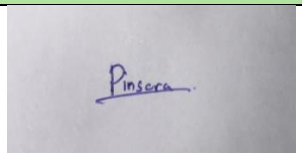
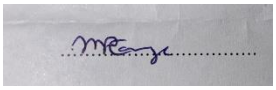
August 2025

Declaration of The Candidate & Supervisor

I declare that this is my work, and this proposal does not incorporate without acknowledgment any material previously submitted for a degree or diploma in any other university or institute of higher learning, and to the best of our knowledge and belief, it does not contain any material previously published or written by another person except where the acknowledgment is made in the text.

Name	Student ID	Signature
PINSARA T H A K	IT22255860	

The above candidates are carrying out research for the undergraduate Dissertation under my supervision.



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Signature of the supervisor Date

Acknowledgment

This paper and the research behind it would not have been possible without the exceptional support of my supervisors, my Research supervisor Miss. Manori Gamage Co-supervisor Mr. Nelum Chathuranga Their enthusiasm, knowledge, and exacting attention to detail have been an inspiration and kept my work on track. Besides my supervisors, I would like to thank the rest of my group members Samoda T W O, D L R Dilochana, Senadheera S C C. for their encouragement, insightful comments, and every other support. And finally, thanks to my parents who always offer support and love.

Abstract

This research aims to address the growing concern of road accidents caused by driver fatigue by introducing a multi-sensor based Drowsiness Detection System. Unlike conventional systems that rely primarily on vision-based methods such as facial recognition or blink detection, this study proposes a holistic approach that integrates both physiological and environmental monitoring to enhance accuracy and reliability. The system leverages a blood pressure and pulse sensor to capture vital physiological signals, while ambient temperature, humidity, light, and noise sensors are deployed to continuously evaluate environmental conditions known to influence driver alertness.

An ESP32 microcontroller serves as the processing unit, fusing sensor inputs into a weighted drowsiness risk scoring model, expressed as Drowsiness Risk Score = X·Physical + Y·Environmental. This dynamic scoring approach ensures that the model adapts in real-time to fluctuating driver states and environmental factors, providing a more context-aware evaluation. When the calculated risk score surpasses a defined threshold, the system instantly triggers a dual-layered alert mechanism: an on-board buzzer for immediate notification and a GSM module that transmits real-time alerts to external stakeholders or mobile devices.

The novelty of this research lies in its pioneering use of multi-sensor fusion, bridging physiological and environmental factors that are often overlooked in conventional systems. This reduces the likelihood of false alarms, improves robustness under varying driving conditions, and provides a cost-effective, scalable framework suitable for real-world vehicular environments. Through accurate detection and timely intervention, the system has the potential to significantly mitigate the risks of fatigue-induced accidents, thereby safeguarding both drivers and passengers.

Ultimately, this research underscores the importance of proactive monitoring in intelligent transportation systems and highlights how sensor-driven decision-making can revolutionize road safety. By shifting from reactive to preventative strategies, this system paves the way for future advancements in smart mobility solutions that prioritize human well-being.

Keywords: Driver drowsiness, Multi-sensor fusion, Physiological monitoring, Environmental sensing, ESP32, GSM alerts, Road safety, Accident prevention.

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1. INTRODUCTION

1.1. Background & Literature Survey

The background and literature survey of this study explore the progression of driver drowsiness detection systems and the emergence of innovative, multi-sensor approaches aimed at reducing fatigue-related accidents. Traditionally, drowsiness detection relied heavily on visual monitoring techniques such as facial recognition, eyelid closure tracking, and yawning frequency measurement. While these camera-based methods provided valuable insights, they often suffered from environmental limitations such as poor lighting conditions, camera obstruction, and reduced accuracy when drivers wore glasses or sunglasses.

Over the past decade, advancements in sensor technology have opened new possibilities for monitoring driver alertness beyond facial cues. Physiological monitoring, such as heart rate and electroencephalogram (EEG) analysis, has shown promise in capturing early signs of fatigue. Environmental monitoring, on the other hand, has revealed that conditions such as high cabin temperature, low ambient light, and persistent noise can significantly accelerate the onset of drowsiness. These findings have shifted the focus towards integrated systems that combine multiple data streams to deliver more reliable and context-aware assessments.

A review of existing literature consistently underscores the importance of early and accurate detection in preventing fatigue-related collisions. Sahayadhas et al. [1] and Lal & Craig [3] emphasize that relying on a single detection modality increases the risk of false positives or false negatives, while multi-factor approaches can significantly improve system robustness. May & Baldwin [2] further highlight that drowsiness is influenced by both physiological and situational factors, and that detection systems must account for these interdependencies to be effective.

Despite the demonstrated potential of combining physiological and environmental parameters, relatively few studies have implemented such integrations in real-world driving environments. Most research efforts have focused on laboratory simulations or have prioritized one factor over the other. This gap presents an opportunity for novel solutions that leverage real-time, embedded sensor systems for continuous monitoring in natural driving conditions.

The integration of physiological monitoring (such as real-time blood pressure) with environmental sensing (temperature, light, and noise levels) represents a paradigm shift in drowsiness detection. While individual studies have validated the importance of each factor independently, the synergistic benefits of their combined application remain underexplored in operational road scenarios.

This literature gap underscores the novelty and significance of the present study, which aims to design and evaluate a multi-sensor fusion model for driver drowsiness risk assessment. By synthesizing prior research findings and identifying unaddressed challenges, this study seeks to contribute to the advancement of intelligent transportation safety systems that proactively mitigate fatigue-related accidents.

1.2. Literature Survey

Here's a literature survey in the same style, tailored to your **multi-sensor drowsiness detection** component:

- **Thiffault & Bergeron (2003)** – Examined individual differences in driver fatigue during monotonous simulated driving. They found that personality traits (e.g., sensation seeking—especially "Experience Seeking") predicted steering variability, a performance marker of fatigue. High-sensation-seeking drivers showed larger steering fluctuations and reported more sleep-at-the-wheel incidents. This supports using simple vehicle/behavioral signals alongside sensors to infer fatigue risk.

- **Sahayadhas et al. (2012)** – Comprehensive review of sensor-based drowsiness detection. They grouped methods into vehicle-based (lane/steering), behavioral (yawn, eye closure, facial cues), and physiological (EEG/ECG/EOG). They argue that effective systems often **combine multiple modalities**, highlighting the need for fusion—directly motivating your blend of physiological (BP/pulse) and environmental inputs.
- **May & Baldwin (2009)** – Discussed causes and detection of driver fatigue, distinguishing **sleep-related** vs **task-related** fatigue. They emphasize that understanding causal factors (circadian timing, workload, monotony) is crucial for countermeasures. This aligns with your design choice to model **context** (temperature, light, noise) rather than relying on a one-size-fits-all signal.
- **Lal & Craig (2001)** – Critical review of the psychophysiology of fatigue. They summarize how **physiological signals** (EEG patterns, HR/HRV, skin conductance) change with drowsiness and provide objective cues for detection. This underpins your inclusion of cardiovascular measures (pulse/BP trends) as part of the **Physical** input to the risk score.
- **Khandoker, Morshed & Mandal (2021)** – Demonstrated **real-time detection** using combined physiological and environmental data. Their findings support embedded implementations that operate on live streams from multiple sensors—strong evidence for your ESP32-based, on-device **weighted risk scoring** pipeline.
- **Horne & Reyner (1995)** – Studied sleep-related crashes in the UK and showed clear **time-of-day peaks** (e.g., 02:00–06:00 and 14:00–16:00), with a high share on monotonous roads and many drivers under 30. This highlights that **environmental/contextual factors** (road type, circadian phase) materially affect risk—supporting your Environmental channel (light, temperature, noise).
- **Rosekind et al. (1994)** – Investigated fatigue in **operational, high-stakes environments** (aviation), showing how performance degrades and safety risks rise. Their work generalizes to transportation, reinforcing the safety case for **proactive monitoring** and timely alerting like your GSM/buzzer-based notifications.

Synthesis / Gap Addressed – Collectively, prior work shows that (i) no single indicator suffices, (ii) physiological measures provide objective drowsiness cues, (iii) context matters (circadian/time-of-day, environment, monotony), and (iv) real-time, embedded detection is feasible. Your component advances this by fusing physical (BP/pulse) and environmental (temperature, light, noise) streams into a weighted, context-aware Drowsiness Risk Score ($X \cdot \text{Physical} + Y \cdot \text{Environmental}$), aiming to reduce false alarms and improve robustness in real-world driving.

1.3. Research Gap

Our research seeks to address a notable gap in the current body of knowledge surrounding driver drowsiness detection systems. While a variety of detection approaches exist — including camera-based facial monitoring, EEG analysis, and environmental sensing — there is a clear lack of integrated solutions that fuse physiological parameters such as blood pressure with environmental conditions like temperature, light, and noise, in a real-time, embedded platform.

This gap is significant because single-modality detection systems are often hindered by situational constraints: camera-based systems fail under poor lighting or obstruction, while environmental monitoring alone cannot directly infer driver physiology. Furthermore, there is insufficient research into context-aware risk scoring algorithms that dynamically adjust detection sensitivity based on combined sensor inputs.

The proposed study aims to bridge this gap by developing a multi-sensor fusion model that operates in real-world driving conditions, integrating both physiological and environmental data into a weighted scoring system for accurate, real-time drowsiness alerts. This work contributes not only to academic discourse but also to the practical deployment of intelligent transportation safety systems capable of reducing fatigue-related road accidents.

Table 1: Comparison with existing system

Reference	Limited integration of physiological, behavioral & environmental indicators	Lack of solutions tailored for Sri Lankan driving conditions	Absence of adaptive weighting mechanisms for sensor reliability	High cost / low affordability for local fleets
[1] Thiffault & Bergeron	Yes	Yes	Yes	No
[2] Sahayadhas et al.	Yes	Yes	Yes	Yes
[3] May & Baldwin	Yes	Yes	Yes	No
[4] Lal & Craig	Yes	Yes	Yes	Yes
[5] Rosekind et al.	Yes	Yes	Yes	No
[6] Horne & Reyner	Yes	Yes	Yes	No

Proposed System	No	No	No	No

In this table:

- "No" indicates that the paper does not address the specified challenge.
- "Yes" indicates that the paper addresses the specified challenge.

1.4. Research Problem

The research at hand is centered on the development of an intelligent, multi-sensor driver drowsiness detection system that integrates physiological monitoring with environmental sensing to provide real-time risk assessment and timely alerts. This undertaking necessitates the creation of a robust detection framework capable of accurately interpreting both physical indicators—such as blood pressure variations—and environmental conditions, including ambient temperature, sunlight intensity, and cabin noise levels. The overarching challenge lies in optimizing the fusion of these diverse data sources into a reliable and context-aware risk scoring algorithm that minimizes false alarms while maintaining high sensitivity to genuine signs of fatigue.

A primary technical hurdle is ensuring that physiological signals are captured and interpreted accurately in real time, despite the potential influence of non-fatigue-related factors such as temporary stress or sudden physical exertion. Similarly, environmental data must be analyzed in

a manner that distinguishes between conditions that genuinely contribute to drowsiness and transient variations that have negligible impact. The complexity of correlating these data streams increases the risk of both false positives, which could reduce driver trust in the system, and false negatives, which could allow dangerous levels of fatigue to go undetected.

Furthermore, the timely delivery of alerts to the driver is crucial for effective intervention. Any delay in detection or alert triggering can significantly reduce the window for corrective action, thereby undermining the system's primary safety objective. This demands an efficient, low-latency embedded processing architecture that can operate seamlessly in real-world driving environments without introducing excessive computational overhead or power consumption.

Addressing these challenges requires a multidisciplinary approach that combines expertise in biomedical sensing, environmental monitoring, embedded systems engineering, and human factors analysis. By refining sensor calibration, enhancing data fusion methodologies, and implementing adaptive weighting mechanisms within the risk scoring algorithm, this research aims to produce a drowsiness detection solution that is not only accurate and reliable but also practical for widespread deployment in modern vehicles.

2. OBJECTIVES

2.1 Main Objective

The primary objective of this research is the development of a Multifactor Driver Drowsiness Risk Evaluation System that leverages advanced sensor technologies and intelligent data fusion algorithms to enhance road safety by accurately detecting early signs of driver fatigue. This system is meticulously designed to address the shortcomings of traditional single-modality detection methods—such as camera-only systems—by integrating both physiological and environmental parameters into a comprehensive, real-time risk assessment framework.

At its core, the proposed system incorporates a real-time blood pressure monitoring module to capture subtle physiological changes that may indicate the onset of fatigue. This is complemented by environmental sensors capable of continuously monitoring ambient temperature, sunlight intensity, and cabin noise levels, all of which are known to influence driver alertness. The integration of these diverse inputs ensures a more robust and context-aware evaluation of driver condition, reducing the likelihood of false positives and false negatives.

The system's architecture is centered around an ESP32 microcontroller, which serves as the central processing unit for data collection, fusion, and risk score computation. By employing a weighted scoring algorithm, the system dynamically adjusts the contribution of each sensor input based on real-time conditions, ensuring that detection sensitivity remains optimal under varying driving environments.

Upon detecting that the computed drowsiness risk score exceeds a predefined safety threshold, the system immediately activates an alert mechanism, which may include audio alarms, visual indicators, or vibration feedback within the driver's seat. This rapid-response capability is critical for providing drivers with sufficient time to take corrective action, thereby preventing potential accidents.

Beyond immediate detection, the system is designed for scalability and integration into modern vehicular safety ecosystems. Its compact hardware footprint and low power consumption make it suitable for integration into both personal vehicles and commercial fleets, enabling widespread adoption without significant infrastructural changes.

In essence, the development of this Multifactor Driver Drowsiness Risk Evaluation System represents a proactive and technologically sophisticated approach to reducing fatigue-related road accidents. By combining precise physiological monitoring with environmental awareness and delivering timely, actionable alerts, the system aspires to set a new standard in intelligent transportation safety solutions.

2.2 Specific Objectives

To achieve the main objective, the specific objectives to be attained in the proposed system are as follows:

[1] **Integration of physiological and environmental sensing modules** – Deploy real-time monitoring of physiological parameters, specifically blood pressure, alongside environmental parameters such as temperature, sunlight intensity, and cabin noise levels to capture a holistic view of driver alertness.

[2] **Implementation of a weighted risk scoring algorithm** – Develop and calibrate an adaptive algorithm that dynamically adjusts the weightage of physiological and environmental inputs, ensuring accurate computation of a drowsiness risk score under varying driving conditions.

[3] **Real-time alert generation upon threshold breach** – Configure the ESP32-based control unit to trigger immediate alerts—via audio alarms, visual signals, or vibration feedback—when the calculated risk score exceeds a defined safety threshold, ensuring timely driver intervention.

[4] **Minimization of false positives and false negatives** – Optimize sensor calibration and algorithm logic to distinguish between genuine fatigue indicators and unrelated environmental or physiological fluctuations, thereby enhancing system reliability and driver trust.

[5] **Validation through real-world vehicle testing** – Conduct structured test scenarios in actual driving environments to evaluate system accuracy, responsiveness, and robustness, and refine operational parameters based on empirical performance data.

3. METHODOLOGY

3.1 Project Overview

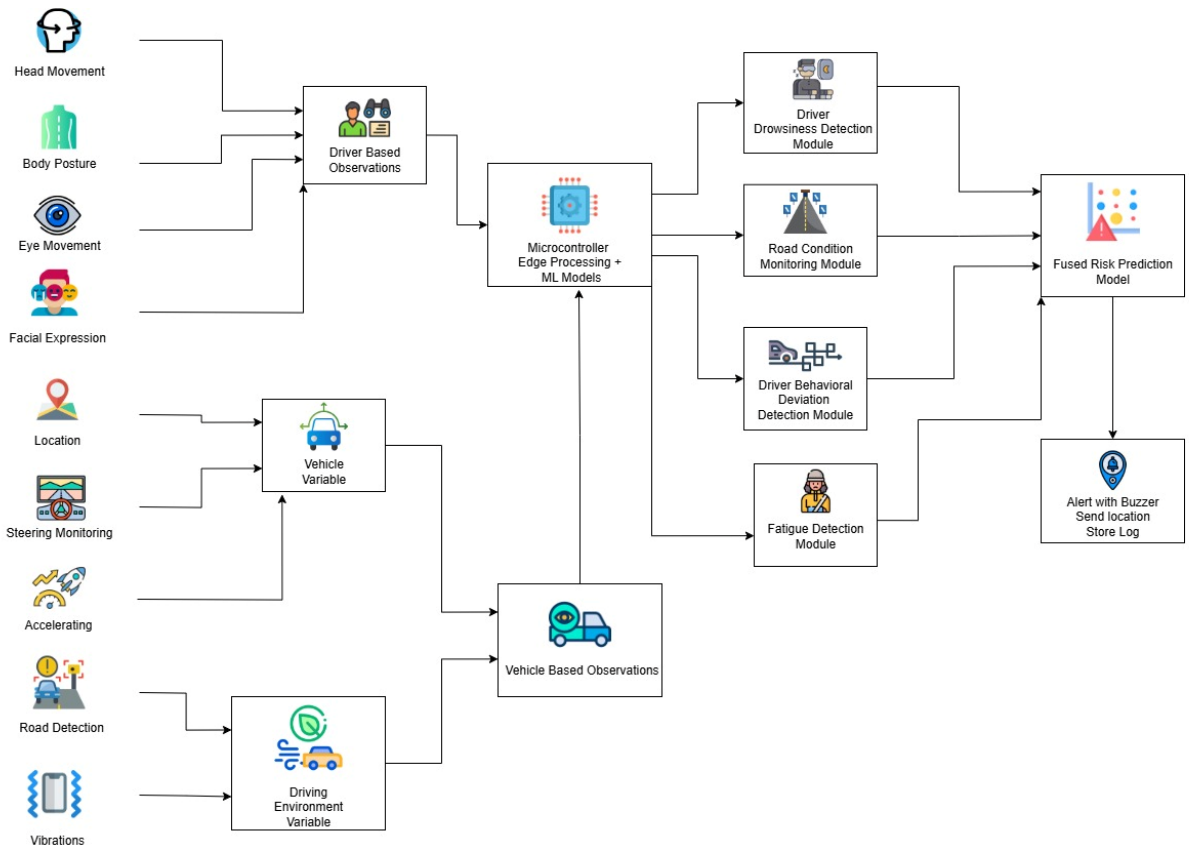


Figure 2: Project Overview Diagram

3.2 System Overview Diagram

The proposed Multifactor Driver Drowsiness Risk Evaluation System introduces an advanced, sensor-driven approach to enhancing road safety by detecting driver fatigue through the integration of physiological and environmental monitoring. At the core of the system lies a blood pressure sensing module that continuously tracks the driver's physiological state, identifying subtle variations that may indicate the onset of drowsiness. Complementing this are environmental sensors—including temperature, light, and noise detectors—which collectively assess conditions within the vehicle cabin that can influence driver alertness.

Sensor readings are processed in real time by an ESP32 microcontroller, which serves as the system's central intelligence unit. This microcontroller employs a weighted risk scoring algorithm that dynamically combines physiological and environmental inputs to calculate a Drowsiness Risk Score. By continuously updating this score, the system ensures accurate, context-aware detection of fatigue while minimizing the occurrence of false alarms.

When the calculated risk score exceeds a predefined safety threshold, the system immediately triggers a driver alert module. This may take the form of audible warnings, dashboard light indicators, or seat vibration feedback, ensuring that the driver is promptly made aware of their compromised alertness. The alert mechanism is designed for minimal latency, guaranteeing a rapid response to potentially dangerous fatigue levels.

Additionally, the system's architecture allows for data logging and analysis, enabling post-test evaluations of sensor readings and detection accuracy. This feature supports both calibration improvements and the fine-tuning of algorithm parameters for different vehicle environments. The modular design ensures adaptability, making it possible to incorporate additional sensing technologies—such as heart rate monitoring or facial recognition—without overhauling the system's core framework.

By integrating continuous physiological monitoring, environmental assessment, and intelligent risk computation into a compact, vehicle-ready platform, this system represents a significant advancement over traditional camera-based fatigue detection methods. The fusion of diverse data streams into a single, actionable risk score ensures that drivers receive timely and accurate alerts, thereby reducing the likelihood of fatigue-induced accidents and contributing to safer roads.

3.3 Individual Component

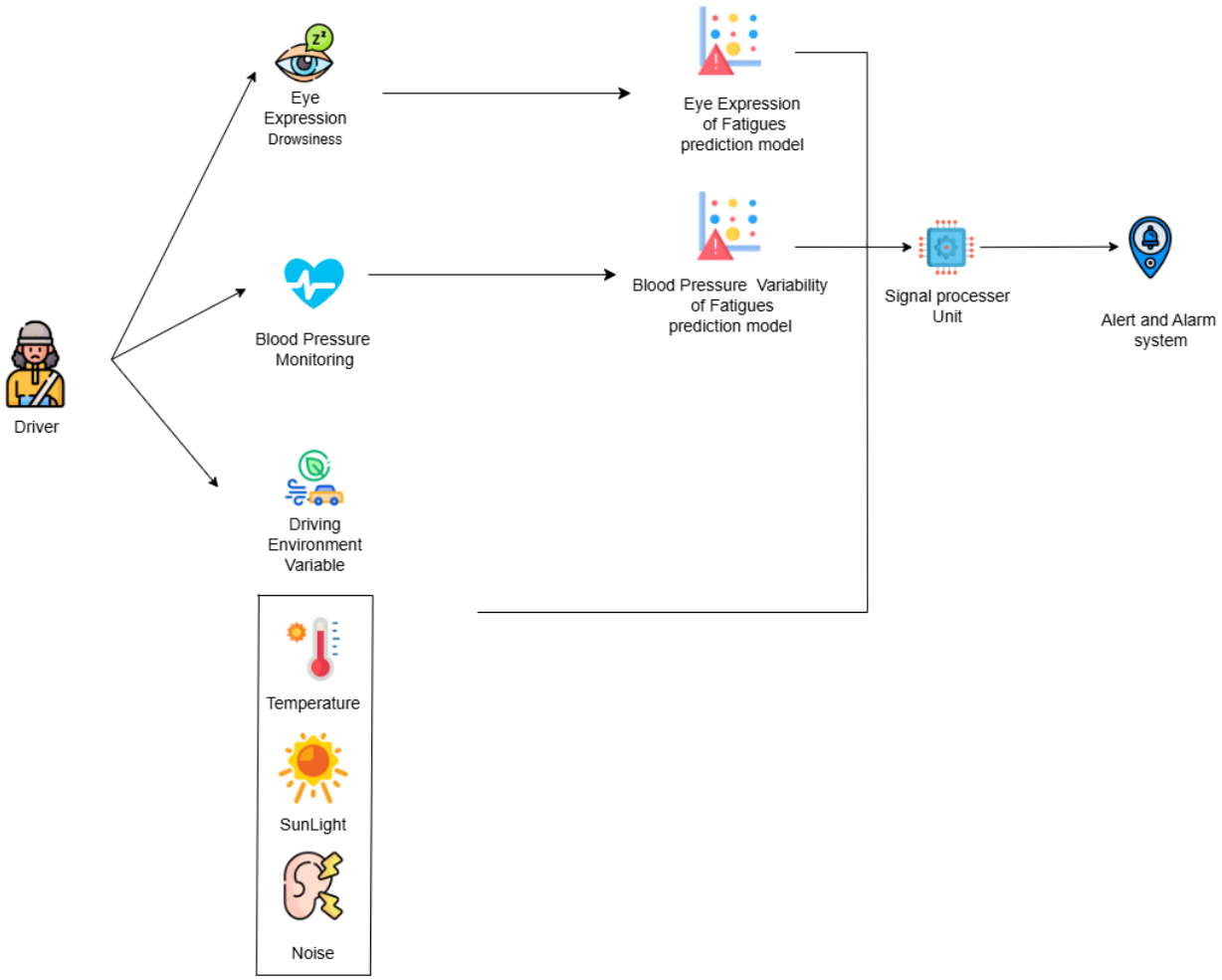


Figure 3: Project Overview Diagram

3.3.2 The flow of the system diagram – Individual Component

The methodology for integrating physiological and environmental sensing into the Multifactor Driver Drowsiness Risk Evaluation System is designed to ensure precise monitoring, accurate risk assessment, and timely alert generation, adhering to formal operational standards.

Initially, the physiological sensing module—comprising a blood pressure sensor—is positioned to unobtrusively monitor the driver's cardiovascular activity in real time. Simultaneously, environmental sensors for temperature, light, and cabin noise are activated to capture relevant surrounding conditions that may contribute to driver fatigue.

These sensor readings are continuously relayed to the ESP32 microcontroller, which functions as the central processing unit. Upon receiving inputs, the ESP32 executes a weighted risk scoring algorithm, designed to evaluate the combined influence of physiological and environmental factors. This algorithm dynamically adjusts the contribution of each factor according to real-time variations, ensuring context-aware analysis.

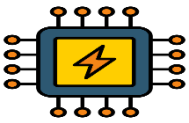
Through this evaluation, the system determines whether the computed Drowsiness Risk Score surpasses a pre-set safety threshold. If the threshold is exceeded, indicating a high probability of driver fatigue, the ESP32 activates the alert system module. This module is capable of delivering immediate warnings via audio alarms, dashboard indicators, or seat vibration feedback to prompt corrective driver action.

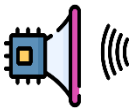
In addition, the system maintains an internal data log of sensor readings and calculated risk scores. This log serves as a reference for further analysis, algorithm refinement, and performance validation during real-world testing scenarios.

By following this structured methodology, the integration of physiological and environmental sensing with an intelligent ESP32-based processing unit enables proactive detection of driver fatigue. This real-time, multi-sensor approach ensures that alerts are both timely and reliable, thereby reducing the likelihood of fatigue-related accidents and enhancing overall road safety.

3.4 Tools and Technologies

Table 2: Tools and Technologies

Tools	Technologies	Explanation
Microcontroller 	ESP32 Board	The ESP32 microcontroller handles sensor data processing and wireless communication. It features integrated Wi-Fi and Bluetooth, making it ideal for IoT applications where data must be exchanged with cloud services or other devices. Its dual-core processor can connect to routers and transmit data, while also performing on-board analysis. In practice, the ESP32 computes the driver's risk score in real time and can issue alerts (e.g. notifications) immediately when a drowsiness threshold is exceeded.
Blood Pressure Sensor	BMP180 + Pulse Module	A pulse sensor measures the driver's heart rate, and the BMP180 (a barometric pressure sensor) provides additional cardiovascular data. These physiological signals are known to change as fatigue sets in. In fact, research shows that heart rate and its variability vary significantly between alert and drowsy states By tracking pulse and pressure, the system gains insight into cardiovascular changes. Sudden drops or abnormal patterns in these measures can indicate drowsiness, so they feed into the fatigue risk model as key indicators.
Temperature & Humidity Sensor	DHT11 or BME680	This sensor pair monitors cabin temperature and humidity to assess thermal comfort. Environmental conditions have a direct impact on driver alertness; for example, high temperature

		and poor humidity can make a driver feel sleepy or uncomfortable. Studies have found that ambient temperature correlates with fatigue symptoms like thirst and tiredness. By measuring these factors, the system can evaluate additional stressors: e.g. if the car interior is very hot or dry, the computed drowsiness risk is adjusted upward to account for the driver's discomfort.
Light Sensor	LDR (Light Dependent Resistor)	The LDR measures ambient light levels inside (or just outside) the vehicle. This helps assess visibility and lighting-related stress. For instance, very low lighting (night or tunnel) can increase fatigue, while sudden bright light (e.g. oncoming headlights) can also distract a driver. In modern vehicles, LDRs are used to automate headlight beams: they detect ambient luminosity and adjust the headlight intensity accordingly. Automated lighting has been shown to reduce driver fatigue. Therefore, by tracking cabin illumination, the system can infer whether poor visibility or glare may be impacting driver alertness.
Noise Sensor 	Sound Detection Module (microphone)	A microphone-based module monitors cabin and ambient sound. It can detect sudden loud noises or specific audio cues that affect concentration. For example, emergency sirens or horn blares are critical signals; systems like Cerence's Emergency Vehicle Detection use built-in cabin microphones to distinguish sirens from normal music and alert the driver. In the drowsiness system, the sound sensor serves to flag unusual acoustic events. If it picks up a sudden noise (or identifies a siren), the system can temporarily raise the alertness score or prompt an immediate warning, since such sounds often require driver attention.
Alert Module	GSM Module and/or Buzzer	This module provides the actual alerts when risk is high. A piezo buzzer (or speaker) can sound an audible alarm inside the car to jolt the driver awake. Simultaneously, a GSM (cellular) module can send an SMS message to preconfigured contacts (or even the driver's phone) if the risk score crosses a danger threshold. Prior implementations of drowsiness systems explicitly use this combination: for example, Kumar et al. connect a buzzer and a GSM modem so that when

		drowsiness is detected, the buzzer sounds and an SMS is sent. This ensures the driver is immediately notified and others are alerted if intervention is needed.
Programming Environment	Arduino IDE	The Arduino IDE is the development platform used to program the ESP32. It allows writing and uploading firmware in C/C++ to the microcontroller. The IDE includes a serial monitor and debugger to test code, and it provides a library manager to easily install drivers for sensors (e.g. BMP or DHT libraries). In practice, the Arduino IDE is the standard environment for ESP32 development, so the project code – which reads all sensors, computes the risk score, and triggers alerts – is developed and debugged within this tool.
Programming Language	C/C++	The system firmware is written in C/C++, providing efficient low-level control of the hardware. C/C++ is the dominant language for embedded microcontrollers because it offers direct memory access and fast execution. In fact, many microcontroller platforms (Arduino, ESP32, STM32, etc.) use C/C++ for exactly these reasons. Using C/C++ lets the program manipulate sensor data registers and communication interfaces directly, which is crucial for real-time processing. This ensures that sensor readings, calculations, and wireless transmissions happen quickly and with minimal overhead.
Dataset	*Driver Drowsiness Dataset (Kaggle)*	For validation, a known dataset of labeled driver images is used. The Kaggle "Driver Drowsiness Dataset (DDD)" contains tens of thousands of cropped driver-face images, each annotated as "drowsy" or "non-drowsy". By comparing the system's sensor-based risk score against this dataset, one can assess accuracy: e.g. if the sensors flag a high fatigue risk when the image label is "drowsy," the system is validated. In essence, this dataset provides ground-truth examples to correlate the multifactor sensor readings with actual drowsiness states observed in drivers.

4. PROJECT REQUIREMENTS

4.1 Functional Requirements

[1] **Sensor Data Acquisition**

- The system must accurately collect real-time physiological data from the blood pressure sensor.
- It must simultaneously acquire environmental data, including temperature, sunlight intensity, and cabin noise levels.
- Sensor readings should be updated continuously to ensure up-to-date monitoring of driver conditions.

[2] **Risk Score Computation**

- The system must implement a weighted risk scoring algorithm that integrates both physiological and environmental parameters.
- The algorithm should dynamically adjust factor weightage based on real-time conditions to improve detection accuracy.
- Risk scores must be computed in milliseconds to facilitate timely decision-making.

[3] **Alert Mechanism**

- When the computed drowsiness risk score exceeds the predefined threshold, the system must immediately activate the alert module.
- Alerts should be delivered through multiple channels, including audio alarms, visual indicators, or vibration feedback.
- The alert mechanism must deactivate once the driver's condition returns to a safe state.

[4] **Data Logging and Analysis**

- The system must maintain an internal database of all collected sensor readings and calculated risk scores.
- Logs should include timestamps, sensor values, and alert events for future analysis and algorithm refinement.

[5] **Component Integration**

- The ESP32 microcontroller must serve as the central hub for data collection, processing, and alert activation.

The system must ensure seamless communication between all sensing modules and the ESP32 to maintain reliable operation.

4.1 Non- Functional Requirements

1] **Performance**

- The system should respond to potential fatigue conditions with minimal latency, ideally within milliseconds.
- Alerts must be delivered in real time to provide drivers with sufficient reaction time.

[2] **Reliability**

- The system must operate reliably under a wide range of environmental conditions, including low light, high temperatures, and varying noise levels.
- It should maintain high accuracy in detecting fatigue, minimizing both false positives and false negatives.

[3] **Scalability**

- The architecture should allow for the integration of additional sensors, such as heart rate monitors or facial recognition modules, without redesigning the system.
- The system should be adaptable for use in various vehicle types, from personal cars to commercial fleets.

[4] **Security**

- All data transmissions between sensors and the ESP32 must be encrypted to prevent unauthorized interception or tampering.
- Access to system configuration and calibration settings must be restricted to authorized personnel.

[5] **Usability**

- The system interface, whether embedded or mobile-based, should be user-friendly, enabling easy setup, monitoring, and configuration.

Comprehensive documentation should be provided to assist with installation, calibration, and maintenance, ensuring user adoption and effective operation.

5. WORK BREAKDOWN CHART

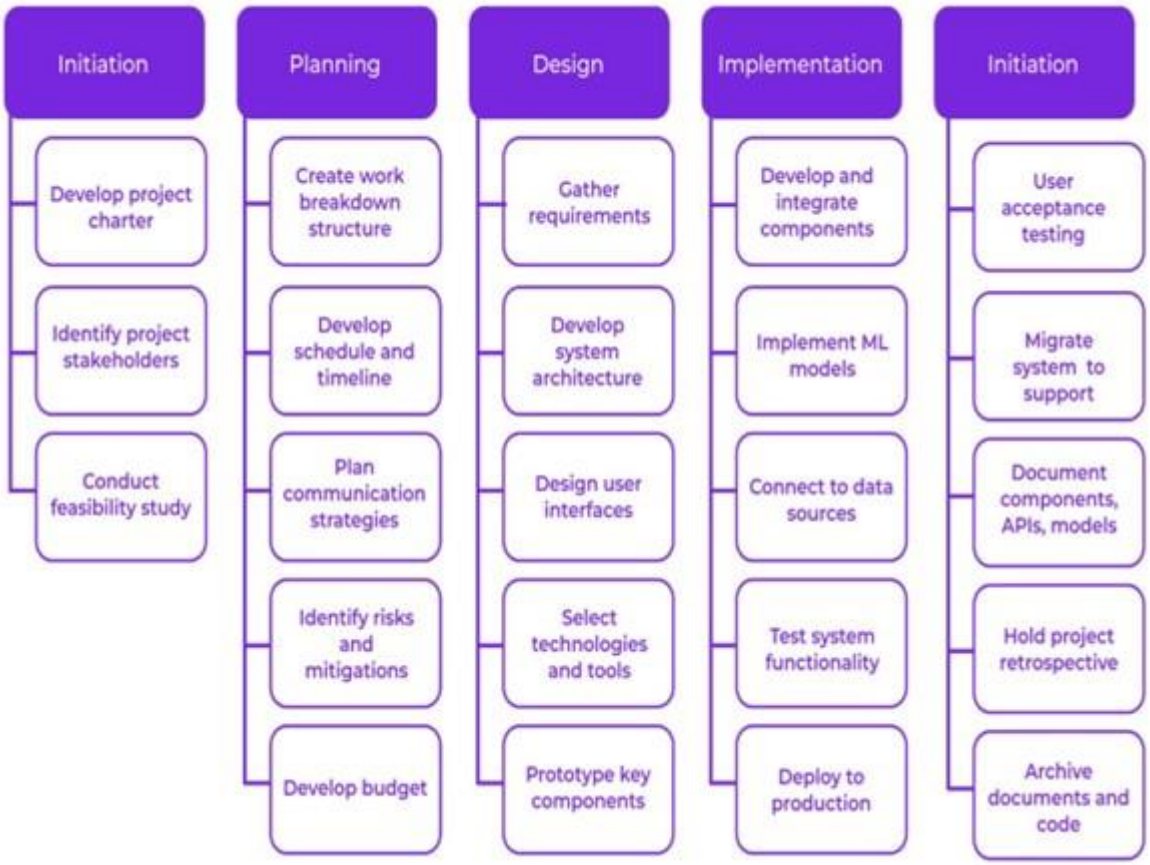
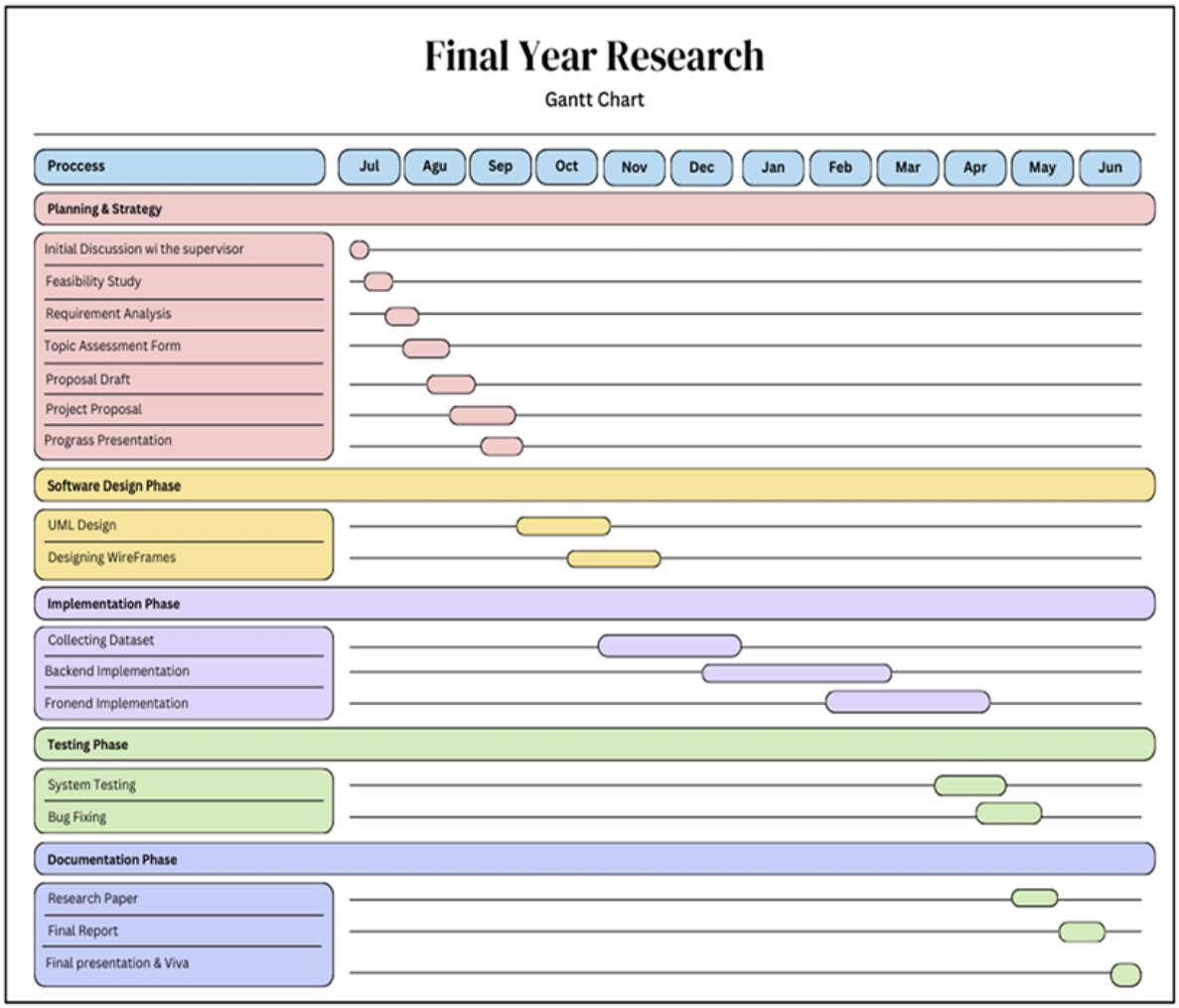


Figure 4: Work breakdown chart

6. GANTT CHART

Table 3: Gantt Chart



7. BUDGET AND BUDGET JUSTIFICATION

The development and implementation of the Drowsiness Detection System requires a careful evaluation of both hardware and software costs. The budget primarily covers the procurement of essential sensors such as blood pressure, temperature, light, and noise sensors, along with the ESP32 microcontroller and alert modules (GSM and buzzer). Additional expenses include accessories such as breadboards, jumper wires, and reliable power supplies to ensure system stability. Since software platforms like the Arduino IDE and datasets from Kaggle are freely available, the cost burden lies mainly on hardware components. Overall, this budget ensures a cost-effective yet comprehensive solution, supporting real-time driver monitoring while maintaining affordability for practical

deployment.*Table 4: Budget and Budget Justification*

Component	Amount
ESP32 microcontroller board	Rs. 1,690
BMP180 sensor + pulse sensor module (blood pressure + pulse)	Rs. 930
DHT11 temperature and humidity sensor	Rs. 270
LDR (light sensor)	Rs. 150
Sound detection module (microphone)	Rs. 150
Buzzer (alert)	Rs. 50
GSM module (SIM800L for SMS alerts)	Rs. 1,090
Breadboard and jumper wires (accessories)	Rs. 640
Arduino IDE (software, free)	Free
Power supply (USB adapter/battery)	Rs. 980

Dataset access (Kaggle, free)	Rs. 0
Total	Rs. 5,950

8. REFERENCES

[1] P. Thiffault and J. Bergeron, “Fatigue and individual differences in monotonous simulated driving,” *Personality and Individual Differences*, vol. 34, no. 1, pp. 159–176, 2003.

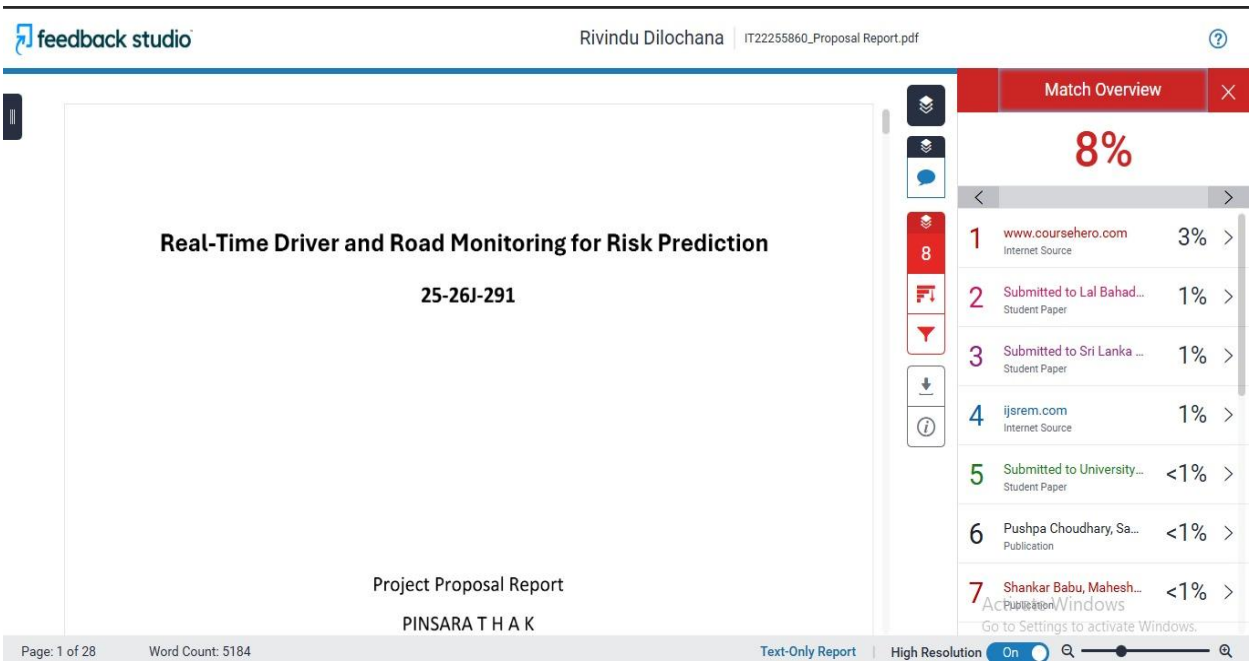
[2] A. Sahayadhas, K. Sundaraj, and M. Murugappan, “Detecting driver drowsiness based on sensors: A review,” *Sensors*, vol. 12, no. 12, pp. 16937–16953, 2012.

[3] J. F. May and C. L. Baldwin, “Driver fatigue: The importance of identifying causal factors of fatigue when considering detection and countermeasure technologies,” *Transp. Res. Part F: Traffic Psychol. and Behav.*, vol. 12, no. 3, pp. 218–224, 2009.

[4] S. K. L. Lal and A. Craig, “A critical review of the psychophysiology of driver fatigue,” *Biological Psychology*, vol. 55, no. 3, pp. 173–194, 2001.

[5] M. R. Rosekind et al., “Fatigue in operational settings: Examples from the aviation environment,” *Human Factors*, vol. 36, no. 2, pp. 327–338, 1994.

[6] J. A. Horne and L. A. Reyner, “Sleep related vehicle accidents,” *BMJ*, vol. 310, no. 6979, pp. 565–567, 1995.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